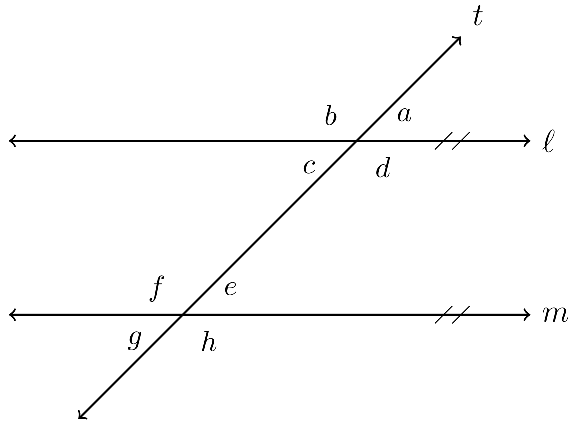
Angle Measures on Parallel Lines

Finding an unknown angle from one that is known, solving for x when a parallel-line angle pair is congruent, and solving for x when the pair adds to 180 degrees

The diagram every problem uses.



Lines ℓ and m are parallel, and t is a transversal cutting both. Every problem on this sheet names its angles from this one diagram. No measures are shown here — use the values given in each problem, and do not read anything off the drawing. (Not to scale.)

How to work these problems.

- Decide first whether the two angles named are **congruent** (corresponding, alternate interior, alternate exterior, or vertical) or **supplementary** (a linear pair, or same-side interior). That decision is the whole problem.
- Congruent means *set the expressions equal.* Supplementary means *add them and set the sum to 180.*
- When a problem gives expressions in x, finding x is only half the job — substitute it back to get the angle the question asked for.
- Name the relationship you used beside each answer.

1. The measure of $\angle a$ is 118°. Find the measures of $\angle d$ and $\angle e$.

$m\angle d =$ ________ $m\angle e =$ ________

2. The measure of $\angle c$ is 47°. Find the measures of $\angle f$ and $\angle g$.

$m\angle f =$ ________ $m\angle g =$ ________

3. $m\angle b = (3x + 10)^\circ$ and $m\angle f = (5x - 24)^\circ$. Find x, then find $m\angle b$.

$x =$ ________ $m\angle b =$ ________

4. The measure of $\angle h$ is 125°. Find the measures of $\angle b$ and $\angle e$.

$m\angle b =$ ________ $m\angle e =$ ________

5. $m\angle d = (7x - 15)^\circ$ and $m\angle f = (4x + 21)^\circ$. Find x, then find $m\angle d$.

$x =$ _________ $m\angle d =$ _________

6. The measure of $\angle f$ is 108°. Find the measures of $\angle d$ and $\angle a$. This one takes two relationships, not one.

$m\angle d =$ _________ $m\angle a =$ _________

7. $m\angle c = (4x + 12)^\circ$ and $m\angle f = (6x + 8)^\circ$. Find x, then find $m\angle c$.

$x =$ _________ $m\angle c =$ _________

8. $m\angle d = (5x - 8)^\circ$ and $m\angle e = (3x + 12)^\circ$. Find x, then find $m\angle d$.

$x =$ _________ $m\angle d =$ _________

9. $m\angle a = (2x + 34)^\circ$ and $m\angle g = (5x - 11)^\circ$. Find x, then find $m\angle a$.

$x =$ _________ $m\angle a =$ _________

10. $m\angle b = (9x - 4)^\circ$ and $m\angle e = (3x + 16)^\circ$. This time nobody tells you which rule applies: decide for yourself whether the two angles are congruent or supplementary, then find x and $m\angle b$. Below your work, write one or two sentences naming the relationship you used and explaining how you knew it was that one rather than the other.

$x =$ __________ $m\angle b =$ __________